



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Driven Grocery Commerce Intelligence & Demand Optimization System

CSA0806 – Python Programming

Name & Register No.

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Dr. Ponmaniraj Sambasivam
Professor

Introduction

- Nowadays, grocery shopping is becoming increasingly complex due to diverse product choices, busy lifestyles, and fluctuating market demand.
- Customers often struggle to find the right products, track their needs, or manage shopping lists efficiently.
- It supports personalized product suggestions, auto-generated shopping lists, and real-time demand prediction.
- This system enhances convenience for users while improving business efficiency and reducing waste for store owners.



Objectives

- ✓ To provide personalized product recommendations based on user preferences, purchase history and behavior patterns.
- ✓ To optimize inventory levels for shop owners by reducing stock-outs and minimizing overstocking.
- ✓ To support data-driven decision-making for both users and store managers using analytics.
- ✓ To integrate all functionalities into a unified system that improves efficiency, reduces waste, and boosts overall customer satisfaction.



Literature Review

S.No.	Author & Year	Focus Area	Key Finding	Relevance
[1]	L. Z. H. Jansen (2024)	Online grocery recommender systems	Challenges and advances in grocery recommender systems.	Supports personalized product recommendation.
[2]	P. Saha (2022)	Retail demand forecasting	Deep learning improves demand forecasting accuracy.	Supports ML-based demand forecasting.
[3]	K. Sadeghi (2024)	Explainable AI and decision-making	Explainable AI enhances transparent and agile decisions.	Enhances trust and transparency in AI predictions.
[4]	W. Villegas-Ch et al. (2024)	Inventory optimization	ML and computer vision optimize inventory and reduce wastage.	Supports inventory optimization and waste reduction.
[5]	E. E. Kosasih (2024)	Explainable AI in supply chain	XAI improves transparency in supply chain decisions.	Promotes transparent and data-driven supply chain decisions.

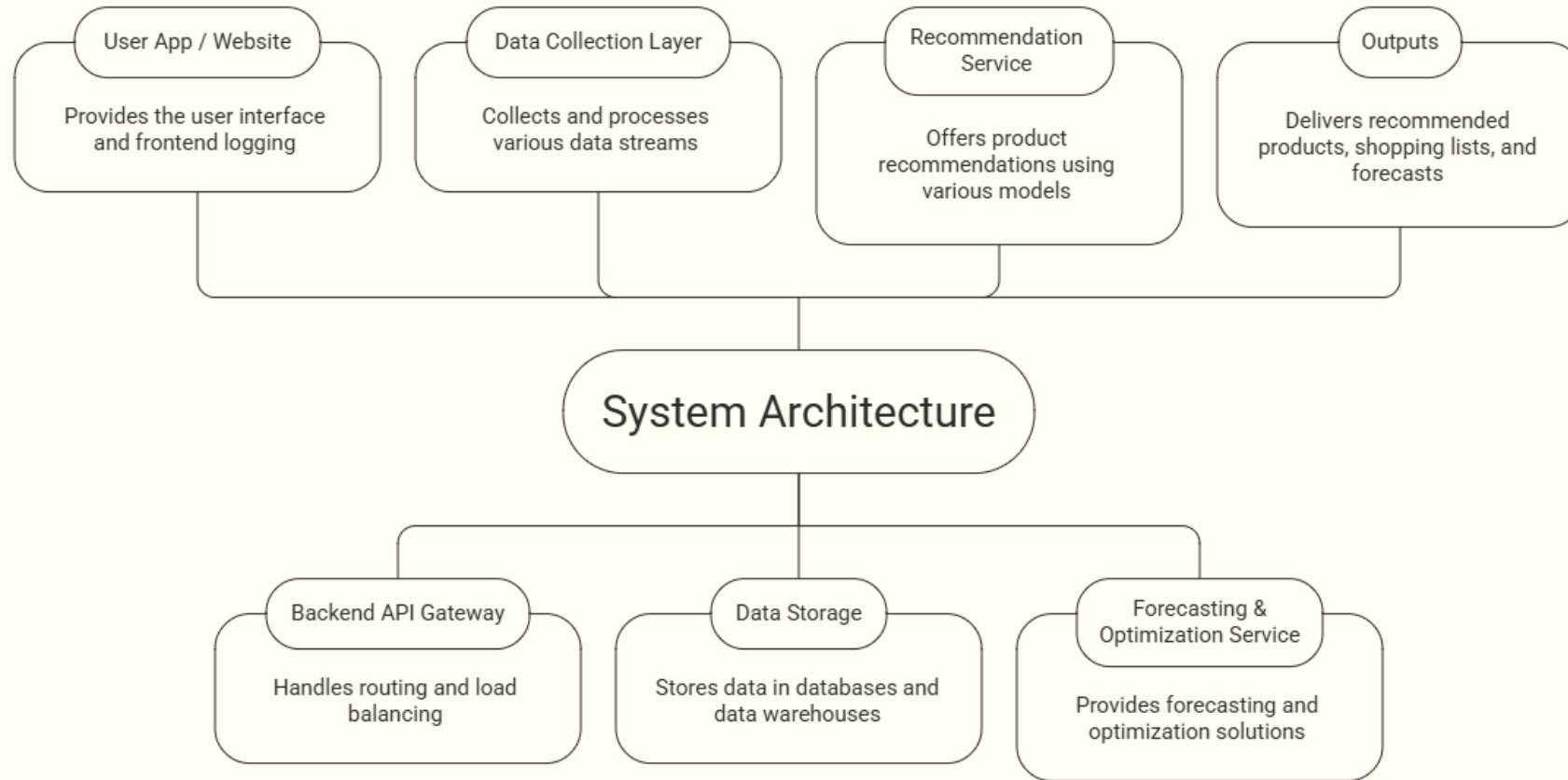


Problem Statement



- ✓ Customers lack personalized product recommendations, making grocery shopping time-consuming and inefficient.
- ✓ Users often forget essential items due to the absence of a smart, predictive shopping list assistant.
- ✓ Manual inventory management causes operational inefficiencies and financial losses.
- ✓ Existing grocery systems do not integrate user behavior analytics with inventory optimization, resulting in poor decision-making.

Architecture Diagram





Technologies Used



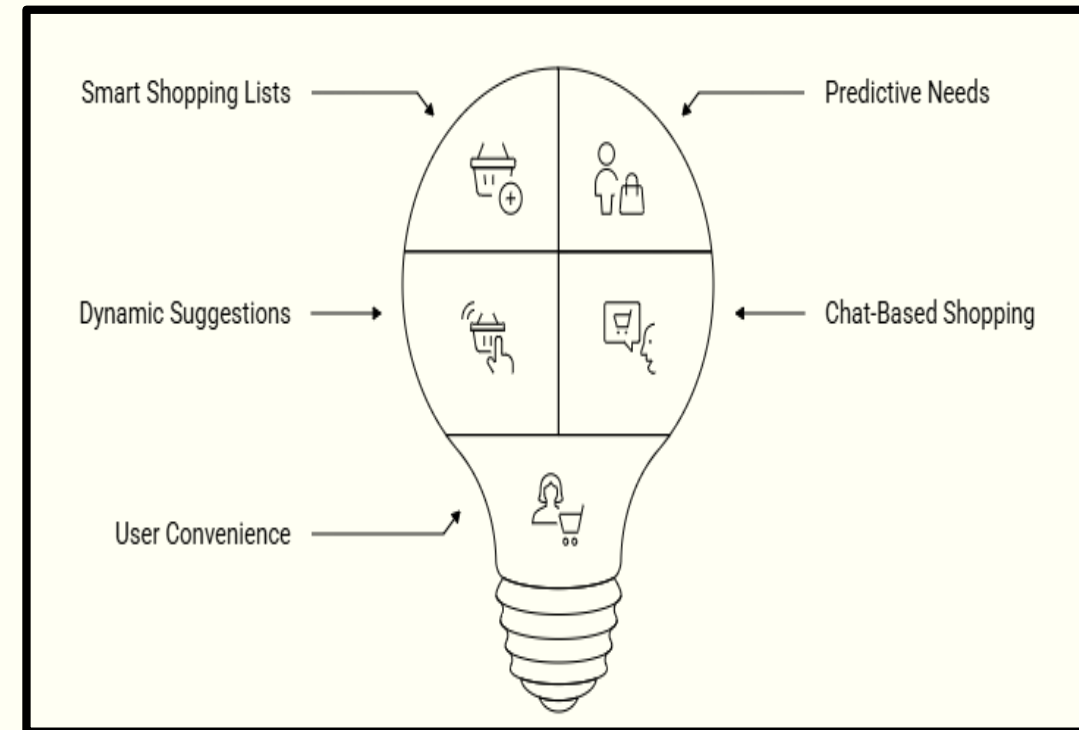
- **Frontend:** HTML, CSS, JavaScript with React.js
- **Backend:** Python with Flask
- **Database:** MySQL stored in Mongodb
- **Machine Learning:** Python libraries such as NumPy, Pandas, Scikit-learn, and Prophet/ARIMA for recommendation and demand forecasting models.
- **Visualization & Deployment:** Matplotlib / Plotly for analytics dashboards and cloud platforms (AWS / GCP) for application deployment



Module 1: Behavioral Commerce Intelligence Engine

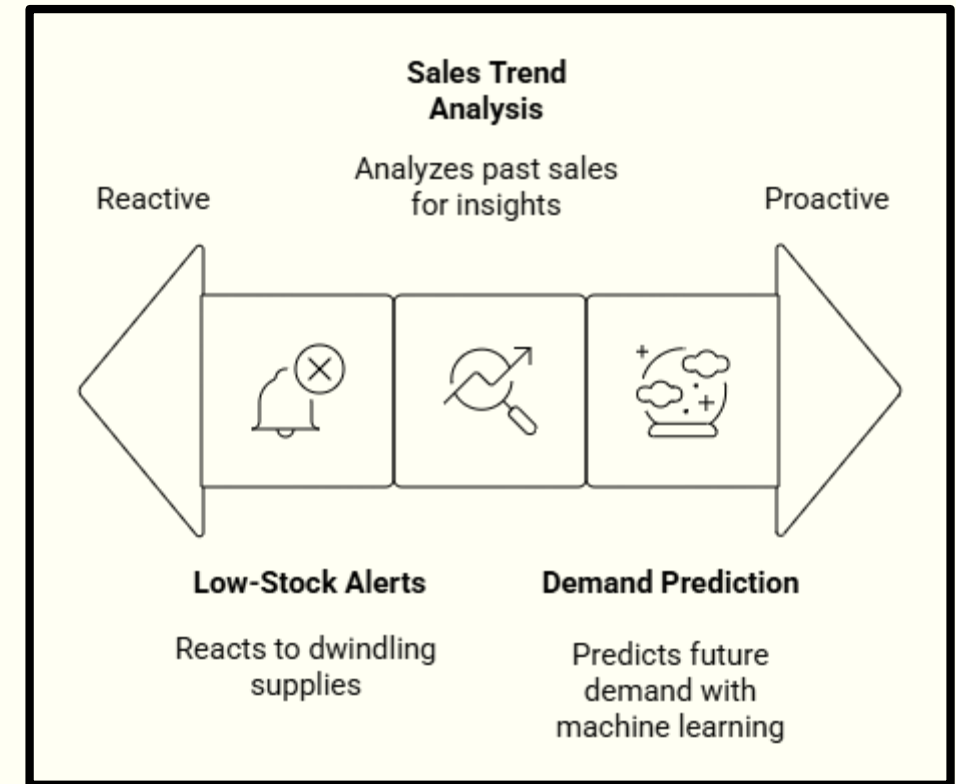


- ✓ Provides AI-driven personalized product recommendations for each user.
- ✓ Automatically generates smart shopping lists based on user purchase patterns.
- ✓ Predicts essential items the user may need soon using behavioral analysis.
- ✓ Offers real-time dynamic product suggestions during browsing.
- ✓ Supports chat-based shopping through NLP assistance.
- ✓ Enhances user convenience by making shopping faster, smarter, and more personalized.



Module 2: Predictive Inventory & Demand Optimization Engine

- ✓ Predicts future product demand using advanced machine learning models.
- ✓ Helps shop owners maintain optimal inventory levels to avoid stock-outs.
- ✓ Reduces overstocking by providing accurate reorder quantity suggestions.
- ✓ Analyzes sales trends, seasonal patterns, and buying behaviors.
- ✓ Generates automated low-stock alerts for timely replenishment.
- ✓ Improves store efficiency with data-driven inventory and supply planning.





Proposed Solution



- ✓ An AI-based system that provides personalized grocery recommendations using user behavior and purchase history.
- ✓ A smart shopping list feature that predicts essential items and reduces missed purchases.
- ✓ Machine learning-based demand forecasting to optimize inventory and minimize stock-related losses.
- ✓ Integration of user analytics with inventory management for data-driven decision-making.
- ✓ Explainable AI techniques to ensure transparency, trust, and effective use of system predictions

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Predict demand, prevent stockouts, and generate cinematic marketing videos instantly.

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GrocerAI

Shop

Shopping List

Admin

Store

Delivery

Sign Out

Fresh Market

Find the best organic produce near you

Search for apples, milk...

AllSpecial OffersFlash DealsProduceDairyBakeryMeatPantryBeverages

VEGAN

Organic Bananas (1 Dozen)

₹60

Sweet, perfectly ripe organic bananas. Great for smoothies or snacking.

Add to Cart

VEGAN

Fresh Avocados (Pack of 3)

₹240

Creamy, nutrient-dense avocados. Perfect for toast or guacamole.

Add to Cart

VEGAN

Red Strawberries (Box)

₹250

Juicy, sweet red strawberries picked at peak ripeness.

Add to Cart

VEGAN

Organic Spinach

₹80

Fresh, vibrant green organic spinach leaves.

Add to Cart

VEGAN

Organic Carrots

₹90

Fresh, crisp organic carrots, perfect for snacking or cooking.

Add to Cart

VEGETARIAN

Whole Milk (1L)

₹72

Fresh whole milk from local farms.

Add to Cart

KR MARKET

InventoryOffersSettings

+ Add Product

PRODUCT	STOCK	PRICE	OFFERS	ACTIONS
Organic Bananas (1 Dozen) Produce	500	₹60	0%	
Fresh Avocados (Pack of 3) Produce	85	₹240	0%	
Red Strawberries (Box) Produce	30	₹250	0%	
Organic Baby Spinach Produce	60	₹80	0%	
Crisp Carrots (1kg) Produce	90	₹90	0%	
Whole Milk (1L) Dairy	42	₹72	0%	
Almond Milk (Unsweetened) Dairy	25	₹299	0%	
Greek Yogurt (400g) Dairy	35	₹250	0%	
Farm Fresh Eggs (Pack of 6) Dairy	50	₹85	0%	
Cheddar Cheese Block Dairy	30	₹340	0%	

Delivery Dashboard

Optimize Route

Current Delivery

Order #8821 In Transit
123 Innovation Dr, Tech City

PICKUP
Fresh Market Store 1
Completed 2:15 PM

DROPOFF
Alex User
Est. 2:45 PM

Mark Delivered

Call Customer

Live Tracking Active

GrocerAI

- Shop
- Shopping List
- Admin
- Store
- Delivery

Sign Out

Platform Analytics

Pending Shop Approvals

No pending approvals.

Total Revenue
₹1,24,592

Active Users
8,540

Avg Order Value
₹450

Sales Trends



Fraud Detection



Potential Fraud Detected

Order #8821 shows unusual location patterns.

Investigate

GrocerAI

Shop

Shopping List

Admin

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Fresh Market

Find the best organic produce near you

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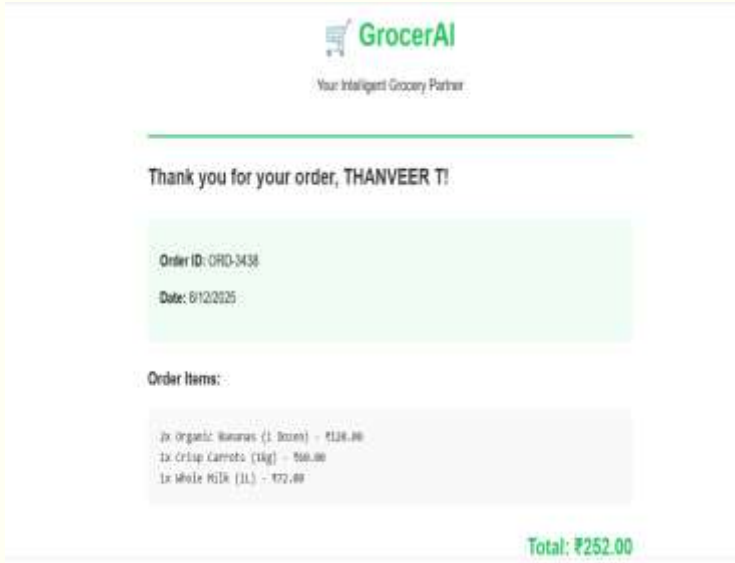
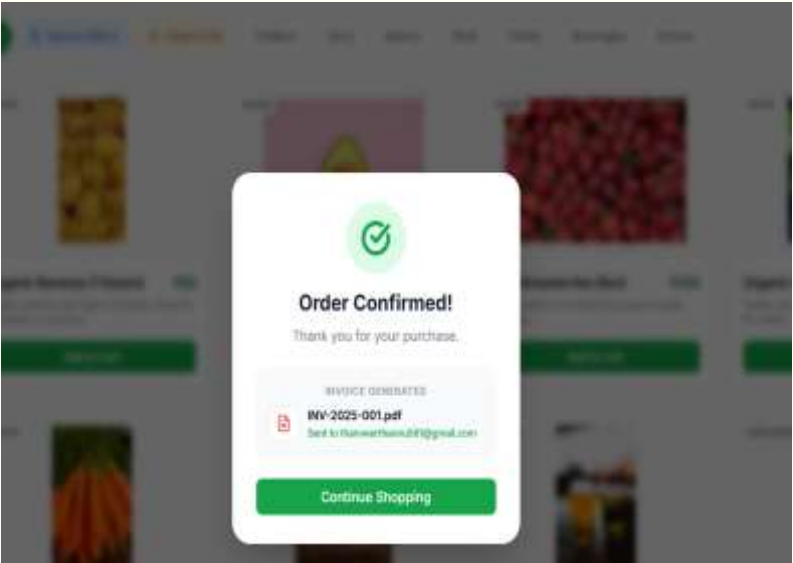
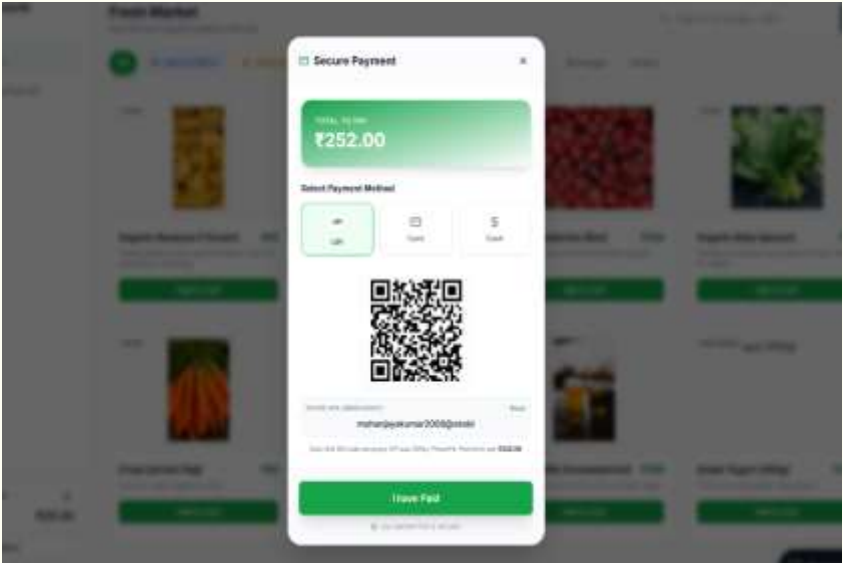
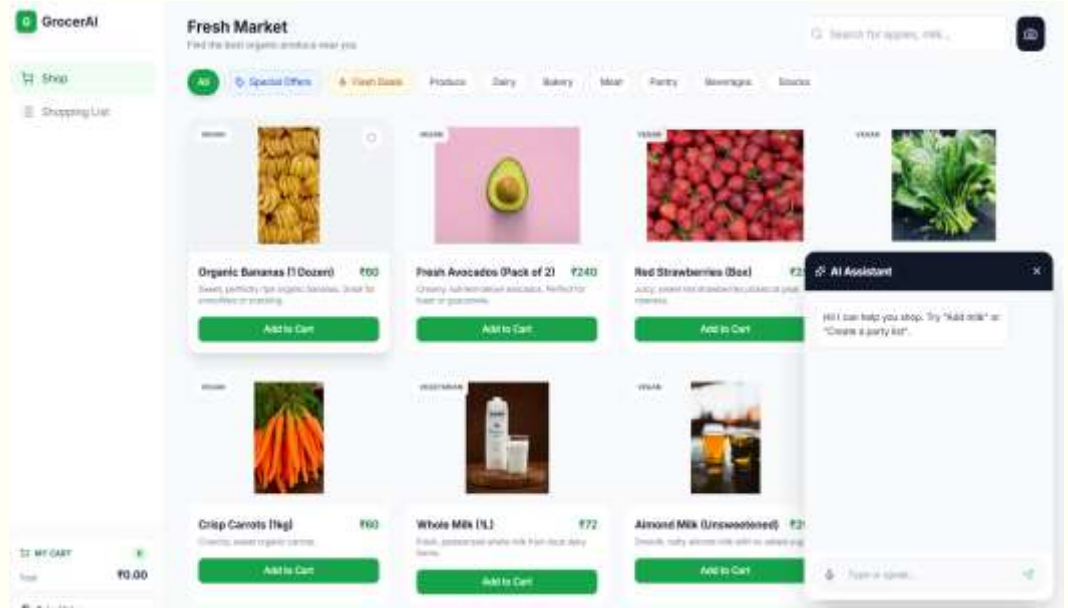
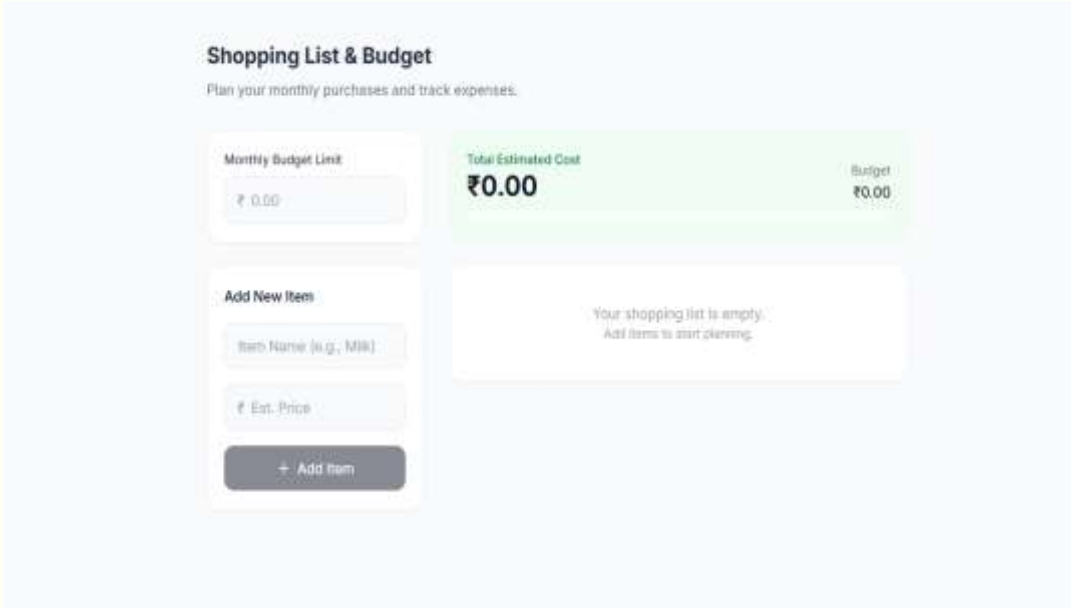
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Engineering Standards



- The project adheres to modular software engineering principles, ensuring maintainable, reusable, and well-structured code across frontend, backend, and machine learning components.
- Data security and privacy standards are implemented using secure authentication mechanisms, role-based access control, and encrypted storage for sensitive user and transaction data.
- Machine learning development standards are followed, including proper data preprocessing, model training, validation, and performance evaluation to ensure reliable predictions.
- The system is designed following scalability and performance engineering standards, allowing it to efficiently handle increasing users, transactions, and data volume.
- User interface and usability standards are applied to deliver a responsive, accessible, and user-friendly experience across all system roles.

ISO/IEC 25010 • ISO/IEC 27001 • ISO/IEC 42001 • IEEE 7000 • IEEE 7001 • ISO 9001

These standards help ensure that the system is **secure, reliable, ethical, transparent, and high-quality.**

Advantages



- ✓ **Smart Cart** – recommend products while shopping.
- ✓ **AI Recommendation** – “Frequently bought together” suggestions.
- ✓ **Auto Shopping List** – generate lists from purchase history.
- ✓ **Budget Assistant** – suggest products within the user's budget.
- ✓ **Demand Forecasting** – predict upcoming product demand.
- ✓ **Inventory Alert** – identify products likely to run out.
- ✓ **Waste Reduction** – help reduce overstock and expired products.
- ✓ **Diet-Aware Recommendation** – recommend products based on dietary preferences.
- ✓ **Sales Dashboard** – show retailers sales and demand trends.

Conclusion

- The proposed system provides a smart, AI-driven solution to improve grocery shopping efficiency and user experience.
- Personalized recommendations and smart shopping lists help users save time and reduce missed purchases.
- Predictive demand forecasting supports better inventory management and minimizes stock-related losses.
- Explainable decision intelligence improves transparency and trust in machine learning outcomes.
- Overall, the system enhances customer satisfaction while enabling data-driven business decisions.



Thank You!!!